

REVIEW

Climate change and diseases of food crops

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Despite complex regional patterns of projected climate change, significant decreases in food crop yields have been predicted using the ‘worst case’ CO₂ emission scenario (A1FI) of the Intergovernmental Panel on Climate Change. Overall, climate change is predicted to have a progressively negative effect on the yield of food crops, particularly in the absence of efforts to mitigate global CO₂ emissions. As with all species, plant pathogens will have varying responses to climate change. Whilst the life cycle of some pathogens will be limited by increasing temperatures, e.g. *Puccinia striiformis* f.sp. *tritici*, other climatic factors such as increasing atmospheric CO₂, may provide more favourable conditions for pathogens such as *Fusarium pseudograminearum*. Based on published literature and unpublished work in progress, we have reviewed the qualitative effects of climate change on pathogens that cause disease of four major food crops: wheat, rice, soybean and potato. The limited data show that the influence will be positive, negative or neutral, depending on the host–pathogen interaction. Quantitative analysis of climate change on pathogens of these crops is largely lacking, either from field or laboratory studies or from modelling-based assessments. Systematic quantitative analysis of these effects will be necessary in developing future disease management plans, such as plant breeding, altered planting schedules, chemical and biological control methods and increased monitoring for new disease threats.

Keywords: climate change, field crop diseases, potato, rice, soybean, wheat

Introduction

The importance of climate to the development of plant diseases has been observed for over 2000 years. The ancient Greeks (370–286 BC) recognized that cereals cultivated at higher altitudes had lower disease incidence than cereals cultivated at lower altitudes (Ghini *et al.*, 2008), but because of the complex nonlinear relationships between a plant and its environment, an assessment of climate-change effects on food cropping systems is not trivial (Semenov, 2009). Analysis of pathogen response to climate in isolation ignores the critical changes that may occur in the host as a result of climate change and the subsequent indirect effects on the pathogen. Increases in leaf waxes and epidermal thickness as a result of increased CO₂ atmospheres (Fuhrer, 2003), for instance, may result in the host having increased physical resistance

to some pathogens. Changes to the architecture of a crop, due to increasing CO₂, may lead to increased humidity within the canopy and more favourable condition for pathogen survival (Chakraborty & Datta, 2003; Pangga *et al.*, 2011). Increased photosynthetic rate under elevated CO₂ levels (Fuhrer, 2003) could lead to the availability of new growth flushes earlier in the season for pathogens to colonize and the subsequent increase in plant biomass will result in a larger reservoir for pathogens to colonize and multiply in. These are just some of the potential indirect effects of rising CO₂ concentration that need to be accounted for when assessing climate change and plant disease.

The major climate-change factors likely to influence plant disease severity and spread include: increased atmospheric CO₂, heavy and unseasonal rains, increased humidity, drought, cyclones and hurricanes and warmer winter temperatures (Cannon, 1998; Chakraborty *et al.*, 2000; Pimentel *et al.*, 2001; Rosenzweig *et al.*, 2001; Berry *et al.*, 2002; Anderson *et al.*, 2004). Changes to any one (or all) of these climatic factors may influence the

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distribution and biology of plant pathogens with positive, negative or neutral effects (Coakley *et al.*, 1999; Fuhrer, 2003).

As an example, the mean temperature in Australia has risen by 0.7°C over the last century, leading to an increase in the number of very warm days and a decrease in frost and cold days. Relative to 1900, the projected increase in annual average temperature varies from 0.4–2.0°C for the year 2030 to 1.0–6.0°C for 2070 (Natural Resource Management Ministerial Council, 2004; CSIRO, BOM, 2007). The Australian continent extends from the tropics to mid-latitudes and covers a wide variety of climates and ecosystems. In mid- to high latitudes, increases in temperatures of 1–3°C, and associated elevated CO₂ and rainfall changes, can have small beneficial impacts on crop yields. In low-latitude regions, even moderate temperature increases of 1–2°C are likely to have negative impacts on yield for major cereals. These results, on the whole, project the potential for global food production to increase with increases in local average temperature over a range of 1–3°C, but above this range they will decrease (Easterling *et al.*, 2007).

The Australian climate is strongly influenced by tropical cyclones and monsoons in Northern Australia and the El Niño-Southern Oscillation (ENSO) phenomenon in Eastern Australia. The ENSO is responsible for significant variations in seasonal to decadal rainfall and can lead to floods and prolonged droughts (Coakley *et al.*, 1999; Pittock *et al.*, 2003). The expected increase in global minimum temperature may enable some plants to start growing earlier in the season, especially in Mediterranean environments in parts of Australia, where growth of winter crops is often limited by low temperatures (Ludwig & Asseng, 2006). The increase in minimum temperature linked to climate change is estimated to be responsible for as much as 30–50% of the observed Australian wheat yield increase (Coakley *et al.*, 1999).

The total annual rainfall over Australia has increased by about 6% over the last hundred years, although this was not homogenous across the continent (Pittock *et al.*, 2003). The future distribution of rainfall is also expected to be spatially and temporarily heterogeneous. Increased precipitation is expected in summer and autumn, mainly in parts of southern and inland Australia, whilst winter and spring are expected to be drier in south-eastern Australia, except Tasmania (Pittock *et al.*, 2003). Projections also suggest potential increases in extreme rainfall in many regions, even where average rainfall may decrease (Natural Resource Management Ministerial Council, 2004).

Changes in rainfall patterns and temperature increases may alter current land use for food crops, resulting in novel plant pathogen or pest problems (Cannon, 1998; Coakley *et al.*, 1999; Parker & Gilbert, 2004) through the introduction of a pest/pathogen in a new crop, which then may infect a previously unaffected indigenous (native or non-native) host; the introduction of a new host, which an indigenous (native or non-native) pest/pathogen may infect; and the introduction

of both new host and new pest/pathogen which may interact with each other. Conversely, climate change may restrict the prevalence of a pest/pathogen or a crop, subsequently restricting the occurrence of disease or pest infestation. Direct, multiple effects on the epidemiology of plant diseases are expected, including the survival of primary inoculum (Melloy *et al.*, 2010), the rate of disease progress during a growing season, and the duration of epidemics.

Rising CO₂ associated with climate change may affect the distribution, abundance and performance of plant insect pests and pathogens (Chakraborty *et al.*, 2008). Increased CO₂ may modify pathogen aggressiveness and/or host susceptibility, affecting the initial establishment of the pathogen on the host (Coakley *et al.*, 1999; Plessl *et al.*, 2005; Matros *et al.*, 2006). Also reported is increased fecundity and growth of some fungal pathogens under elevated CO₂ (Hibberd *et al.*, 1996; Coakley *et al.*, 1999; Chakraborty *et al.*, 2000). Together, an increase in plant canopy size (especially in combination with humidity), and an increase in host abundance and biomass can increase the size of pathogen populations (Manning & Tiedemann, 1995; Chakraborty & Datta, 2003; Mitchell *et al.*, 2003; Pangga *et al.*, 2004).

In agriculture, plant breeding programmes are expected to adapt many crops to increased duration of growing seasons and, concurrently, to develop drought and temperature stress tolerance. There will be opportunities for new cultivars to be introduced, but effective disease screening systems must be in place to prevent pathogens from being introduced with these new crops (Boland *et al.*, 2004). The adaptation of pathogens to a changing climate may prove to be one of the most important challenges under future climates (Garrett *et al.*, 2006), especially in terms of managing the productivity and sustainability of food crops.

The aim of this paper is to summarize research on the influence of climate change on diseases of food crops, enabling the identification of future weaknesses in the pathogens that may be exploited in designing management strategies. This paper complements assessments of climate-change impacts on diseases of forest trees (Sturrock *et al.*, 2011), tropical and plantation crops (Ghini *et al.*, 2011).

Climate change and diseases of food crops

This paper focuses on four food crops grown globally: wheat, rice, soybean and potatoes, and reviews the effects of climate change on key diseases affecting their production.

Wheat diseases

In the US, predicted negative effects on yield of crops such as wheat, maize and barley have implications for the continued production of food for the entire world. A decrease in crop yield by 25–44% under the slowest warming projections and 60–79% under the most rapid

warming projections (Lobell & Field, 2007) will significantly diminish this supply. In China, models predict that without CO₂ fertilization (derived from increasing atmospheric CO₂) enhancing biomass and yield of crops, climate change could reduce rice, maize and wheat yields by up to 37% in the next 20–80 years (Erda *et al.*, 2005).

Rust diseases, such as wheat stripe rust caused by *Puccinia striiformis* f.sp. *tritici* (Pst), are serious pathogens of cereals around the world. Moisture, temperature and wind are the three most important weather factors affecting epidemics of Pst. Increasing global temperatures may limit the development and survival of Pst in some wheat-growing regions around the world. In the laboratory, the optimum constant temperature for urediniospore germination in *P. striiformis* was reported as 10–13°C and the maximum temperature 20–26°C (Tollenaar, 1985), although the disease was shown to flourish under field conditions at 19–30°C (Park, 1990). For example, in China, the prevalence of Pst decreased in association with increase in average annual temperature from 1950 to 1995 (Yang *et al.*, 1998). Similarly, increased frequency and severity of Pst epidemics was associated with increased winter temperatures and lower spring temperatures in the Pacific Northwest in the USA (Coakley, 1979). In contrast, new isolates of Pst, better adapted to high temperatures, now dominate the pathogen population in south central USA (Milus *et al.*, 2006, 2009). Another example of pathogen response to temperature was demonstrated in the UK, where approximately half of the wheat cultivars used showed differential resistance expression when tested against isolates of brown rust (leaf rust) (caused by *Puccinia triticina*), either effective at 10°C and not at 25°C, or vice versa (Jones & Clifford, 1997; Jones, 2000). The effect was not necessarily attributable simply to resistance gene expression response to temperature, as fungal isolates also showed differential temperature responses independent of resistance responses.

Stripe rust infection has also been the focus of modeling studies to assess the potential impact of climate change. White *et al.* (2004) developed process-based models linking life cycles of stripe rust and its host, wheat, to help formulate disease management strategies. Chakraborty *et al.* (2002) assessed climate-change scenarios with stripe rust levels in terms of temperature, rainfall and humidity changes in Australia. There was a clear location effect, making it difficult to generalize, but the overall stripe rust levels at many sites were predicted to be higher for 2070.

Resistant wheat varieties are used to control the development of disease, although Pst can evolve and overcome resistance genes. For example, Pst was monitored in annual surveys in Australia from the first recording in 1979 and 15 pathotypes were detected in wheat in Australia and New Zealand during the 10-year period to 1988 (Wellings & McIntosh, 1990). It is not known how a changing climate will influence the ability of the pathogen to overcome resistant cultivars. If temperature increases do not limit the development of Pst in wheat,

the effect of increasing CO₂ could increase plant biomass, thereby increasing the area susceptible for Pst infection and Pst populations. One study suggests that predicted climate change may result in reduced incidence of wheat stripe rust in future climates for some regions and cultivars and a trend towards lower disease levels in El Niño years (Chakraborty *et al.*, 2002). In a recent review, Chakraborty *et al.* (2010) gave increased crop losses from rusts, faster evolution of new rust races and reduced effectiveness of rust resistance as the three major risks arising from climate-change influence on wheat rusts.

Necrotrophic pathogens have always been a major constraint to wheat production worldwide, but unlike rusts, internationally coordinated research efforts are not common for these pathogens. In Australia, of the three most economically important pathogens, *Pyrenophora tritici-repentis*, *Puccinia striiformis* and *Phaeosphaeria nodorum*, two are necrotrophs (Murray & Brennan, 2009). Other important diseases, such as fusarium head blight and crown rot of wheat caused by several *Fusarium* species, have re-emerged with the adoption of stubble retention practices. A comprehensive analysis of climate-change influence on fusarium head blight and crown rot affecting yield and quality of wheat appears in Chakraborty & Newton (2011). Crown rot caused by *Fusarium pseudograminearum* has been prevalent in southern Australia during the last 12 years of drought where soil moisture has been low. The decreases in mean precipitation projected for winter and spring for the southern Australian wheat belt indicate continued drought conditions which are likely to result in crown rot remaining an important disease for wheat growers. Under elevated CO₂ (825 p.p.m.), fungal biomass significantly increased in the wheat stems compared to infected wheat grown at ambient CO₂ in a relative comparison of fungal DNA to wheat DNA using quantitative RT-PCR (Melloy *et al.*, 2010). The production and survival of crown rot inoculum in stubble will increase as a consequence. The continued drought and warming conditions indicate that crown rot will remain an important disease for Australian wheat growers (Spackman *et al.*, 2008). Among other necrotrophic pathogens, severity of spot blotch caused by *Cochliobolus sativus* has increased in South Asia with higher average night-time temperatures (Sharma *et al.*, 2007).

Yellow dwarf viruses (YDVs) cause some of the most serious diseases of cereal crops worldwide (Lister & Ranieri, 1995). They have recently been classified into two groups: barley yellow dwarf viruses (BYDV) (*Luteoviridae*: *Luteovirus*) and cereal yellow dwarf viruses (CYDV) (*Luteoviridae*: *Polerovirus*) (Mayo & D'Arcy, 1999), the strains of which are distinguished by distinct genome properties and their primary aphid vectors (Lapierre & Signoret, 2004; Katis *et al.*, 2007).

Outbreaks of YDVs are likely to be worst in years when wet cool summers allow larger than normal numbers of aphids and alternate hosts to survive summer. The predicted changes in rainfall in south-eastern Australia are likely to favour the early buildup of aphid populations,

resulting in more frequent and severe YDV infections. Increased autumn rain predicted for some regions and milder winters would allow greater aphid activity and therefore an increase in the incidence and severity of disease (Chakraborty *et al.*, 1998). Some degree of control of YDVs can be achieved by monitoring and spraying for aphids early in the season, but resistance breeding is the most effective solution (Hollaway & Bedggood, 2003). YDV infection of oats (*Avena sativa*) under elevated CO₂ can lead to increased biomass compared to uninfected oat plants, potentially increasing the aphid infestation and virus reservoir (Malmström & Field, 1997).

The response of insect herbivores to temperature have been reviewed by Bale (2002) and found to be the predominant influence on development, survival, range and abundance of herbivore populations, including aphids. Also important are changes mediated through the host plant. Elevated CO₂ can change the nutritional quality of the plant leading to alterations in herbivore feeding behaviour, growth rates, fecundity and population density (Awmack *et al.*, 1996; Whittaker, 1999; Fuhrer, 2003; Stiling & Cornelissen, 2007). Vegetation range shifts and increases in the length of the growing season can be the result of altered temperature and rainfall patterns (Parmesan & Yohe, 2003), which will influence host plant availability and phenologies. Any influences on the vector will in turn impact on the incidence and severity of any potential YDV infection.

Rice diseases

Rice is a staple food crop of the world's population. The demand for this crop continues to grow steadily as the population increases, whilst land and water resources are on the decline (Nguyen, 2002). Increasing temperature, rising sea levels and changes in patterns of rainfall will most likely lead to substantial modifications in land and water resources for rice production (Nguyen, 2002).

Rice blast (*Magnaporthe grisea/oryzae* complex) and sheath blight (*Rhizoctonia solani*) causes crop losses annually worldwide. Rice blast severity was predicted to increase for most of the simulated scenarios using a CERES-Rice model coupled with BLASTSIM (Luo *et al.*, 1998). Temperature change emerged as the main determinant of crop loss; in the cool subtropics such as Japan and northern China, an increase above normal temperature resulted in more severe blast epidemics, whilst in the warm/cool humid subtropics, temperature increase caused significantly fewer blast epidemics (Luo *et al.*, 1998). Rising temperature can also boost effectiveness of some resistance genes, such as Xa7, which is more effective against *Xanthomonas oryzae* pv. *oryzae* at high temperatures (Webb *et al.*, 2010).

In the rice free air CO₂ enrichment (FACE, Okada *et al.*, 2001) experiment in Shizukuishi, northern Japan, rice plants were grown under elevated CO₂ to primarily assess yield effects under future climates. For the first time, plant diseases were studied in open facilities under enriched atmospheric CO₂ environments over three

seasons (1998–2000). The first pathogen assessed was *M. oryzae*, with a pathosystem that comprises two elements: leaf blast and panicle blast. On rice grown under elevated CO₂, leaf blast severity was significantly higher than on rice grown under ambient CO₂. However, panicle blast severity was unchanged when artificially inoculated, but slightly higher under natural infection conditions (Kobayashi *et al.*, 2006).

When *R. solani* was inoculated onto susceptible rice plants in the same experiments, the percentage of diseased plants was significantly higher under elevated CO₂, but the size of lesions was unaltered (Kobayashi *et al.*, 2006). The authors postulated that the increase in leaf blast and sheath blight severity was caused by the lower leaf silicon content in rice grown under elevated CO₂, which may have contributed to the susceptibility of the rice plants.

Soybean diseases

As one of the world's major and fastest-expanding crops, soybean contributes significantly to overall human nutrition in terms of both calorie and protein intake. Soybean is an oilseed crop grown for a variety of uses, e.g. vegetable oils, processed foods such as tofu and soymilk, and soy meal for animal feed. Cultivation is highly concentrated geographically, with only four main producers – USA, Brazil, Argentina and China accounting for almost 90% of world output. The effects of climate change on soybean production have been simulated based on studies under elevated CO₂ in SoyFACE (Ainsworth & Long, 2005; Ainsworth *et al.*, 2008).

Eastburn *et al.* (2010) evaluated the effects of elevated CO₂ and ozone (O₃) on downy mildew, septoria brown spot and sudden death syndrome (SDS) in SoyFACE. Elevated CO₂ alone or in combination with elevated O₃ significantly reduced downy mildew disease severity by 39–66%. In contrast, the same conditions significantly increased brown spot severity, but the increase was small. The atmospheric treatments had no effect on the incidence of SDS or brown spot over three seasons.

Phakospora pachyrhizi causing Asian soybean rust (ASR) is an exceptionally aggressive species that has spread rapidly over the past decade, moving from Asia through Africa on to South America. In September 2004, hurricane Ivan struck the US Gulf Coast, introducing ASR into Louisiana, and by 3 December 2004 there were 20 new detections in eight states, outlining the periphery of the hurricane impact zone (Isard *et al.*, 2005).

In another example, hurricane Wilma (2005) spread *Xanthomonas citri* subsp. *citri* through citrus orchards in Florida, to a point where the US Department of Agriculture declared that containment was no longer an economically feasible option. Wind speeds and driving rain associated with Wilma had not been accounted for in the citrus canker quarantine zone calculations (Irey *et al.*, 2006; Gottwald & Irey, 2007). An increased frequency of severe storm events such as Wilma and Ivan may see the increased dispersal of airborne plant pathogens such as

rusts, splash-borne pathogens such as bacteria, and wind-borne insects and vectors such as aphids and psyllids.

There is observational evidence of an increase in intense tropical cyclone activity in the North Atlantic since about 1970 (Pachauri & Reisinger, 2007), but only suggestions of increased intense tropical cyclone activity in other regions where concerns over data quality are greater. Multi-decadal variability and the quality of the tropical cyclone records prior to routine satellite observations (approx. 1970) complicate the detection of long-term trends in tropical cyclone activity (Pachauri & Reisinger, 2007). Although increased cyclone activity on a global scale is less certain than other climate-change predictions, severe storm events have the ability to spread plant pathogens through damaging plant parts and dispersing cells through wind and driving rain.

Potato diseases

Potatoes are important for the diets and livelihoods of millions of people worldwide. Global potato production has grown markedly in the past years and in 2005, for the first time, more potatoes were grown in developing countries than in industrialized nations. The main producer is China, with a crop yield of 71 million tonnes, which amounts to over 20% of global production (Stäubli *et al.*, 2008).

Without adaptation measures, it is predicted that potato yield will decrease by 18–32% with climate change.

At high latitudes, incremental warming will most likely lead to changes in the time of planting, the use of later-maturing cultivars, and a shift of the location of potato production (Hijmans, 2003). Higher temperatures in growing areas at higher latitudes will lead to a longer growing season and this will in turn lead to increased pest and disease pressure (Haverkort & Verhagen, 2008).

The effects of climate change on five diseases caused by fungi/oomycetes affecting potato in Canada were published by Boland *et al.* (2004). In this study, which included the assessment of inoculum levels, disease establishment, disease progress and duration of epidemic, only necrotrophic *Verticillium* spp. (Table 1), were predicted to increase as a result of climate change (primarily because of increasing temperature). Whilst the primary inoculum levels and duration of *P. infestans* epidemics would increase, Boland *et al.* (2004) predicted that the rate of disease progress would decrease.

In a historical evaluation of climate-change trends on late blight in Finland between 1933 and 2002, and in contrast to the Canadian assessment, the risk of potato late blight outbreak was 17-fold higher during the period 1998–2002 than the periods 1933–62 and 1983–97. Simultaneously, it was quantified that the outbreaks of the epidemics began 2–4 weeks earlier during 1998–2002. The authors associated these observed trends with a climate more conducive to blight in the late 1990s (Hanukkala *et al.*, 2007). Furthermore, simulation modelling found that an increase in mean temperature in southern

Finland of 1°C will extend the late blight infection period by 10–20 days (Kaukoranta, 1996).

Of the three bacterial diseases of potato, only *Streptomyces scabies* was likely to increase in prevalence in Canada (Boland *et al.*, 2004). A net positive increase in prevalence of *Potato leafroll virus* was seen in this study as a reflection of the importance of increased vector populations resulting from warmer winter temperatures (Table 1).

Haverkort & Verhagen (2008) suggested that higher temperatures and irregular rainfall (with the risk of flooding) increases the risks of bacterial infections of potato such as *Pectobacterium carotovorum* var. *carotovorum* and *P. chrysanthemi* (*Dickeya chrysanthemi*) and the increased spread of *Ralstonia solanacearum*.

Knowledge gaps and future research

We have reviewed climate-change-related research on diseases of four important crops, grown for food or oil on large scales around the globe. Overall, there has been limited research on diseases and pathogens of broadacre crops. Most assessments have used existing data on the influence of environmental factors such as temperature to project potential climate-change impacts and only three have used empirical research under realistic field conditions. However, none has used combinations of changing climatic and atmospheric variables in its projection, for instance, combining the influence of increasing temperature with rising CO₂. These limited findings show that diseases caused by necrotrophic or biotrophic pathogens may increase or decrease, or severity may remain unchanged (Table 1). In some instances, a disease has been projected to increase or decrease depending on the region and/or the weather variables used. A regional bias is to be expected given that projections of changes in climatic factors including temperature and rainfall vary according to geographical regions. Given the overall paucity of knowledge and the variability associated with the fragmented studies, it is not possible to generalize about how diseases of food crops may behave under a changing climate.

Given the number of interacting factors determining disease outcomes, modelling approaches will offer the best approach for developing realistic projections (Scherer, 2004). Models used to predict the likely effects of climate change on a pathogen's distribution or biology are increasingly able to account for the complex interactions between a pathogen, its host and the environment.

Despite this, there are critical gaps resulting from a fundamental lack of species data (both spatial and temporal) at a field scale and at the cellular and genomic levels. Measuring how a pathogen is affected by climate change at a cellular or genetic level, such as at elevated CO₂ levels (Matros *et al.*, 2006; Lake & Wade, 2009) will provide insight into the prevalence of the disease in future climates. The challenge will be in linking this data to host–pathogen interactions on a spatial scale in order to determine future management options. Detailed

Table 1 Summary of the effects of selected climate change parameters on diseases of field crops

Crop	Disease/pathogen	Pathogen mode of nutrition	Summary of influence	Overall assessment	Reference
Barley	Powdery mildew – <i>Blumeria graminis</i>	Biotrophic	Decrease at high CO ₂	Decrease	Hibberd <i>et al.</i> (1996)
Barley	Barley yellow dwarf virus	Biotrophic	Decrease at high CO ₂	Decrease	Malmström & Field (1997)
Barley	Smut – <i>Ustilago hordei</i>	Biotrophic	Increased smutted ears at high CO ₂	Increase	Manning & Tiedemann (1995)
Maize	Smut – <i>Ustilago maydis</i>	Biotrophic	Inhibited at high CO ₂	Decrease	Manning & Tiedemann (1995)
Potato	Potato leafroll virus	Biotrophic	Increase	Increase	Boland <i>et al.</i> (2004)
Potato	Six necrotrophic pathogens	Necrotrophic	Decrease	Decrease	Boland <i>et al.</i> (2004)
Potato	Two necrotrophic pathogens	Necrotrophic	Increase	Increase	Boland <i>et al.</i> (2004)
Potato	Late blight – <i>Phytophthora infestans</i>	Necrotrophic	Historically earlier start of more severe epidemics with changing climate	Increase	Hannukkala <i>et al.</i> (2007)
Potato	Early blight – <i>Alternaria solani</i>	Necrotrophic	No effect of high CO ₂	No change	Manning & Tiedemann (1995)
Rice	Leaf blast – <i>Magnaporthe oryzae</i>	Necrotrophic	Increased at high CO ₂	Increase	Kobayashi <i>et al.</i> (2006)
Rice	Sheath blight – <i>Rhizoctonia solani</i>	Necrotrophic	Increased at high CO ₂	Increase	Kobayashi <i>et al.</i> (2006)
Soybean	Downy mildew – <i>Peronospora manshurica</i>	Biotrophic	Decrease	Decrease	Boland <i>et al.</i> (2004)
Soybean	Downy mildew – <i>Peronospora manshurica</i>	Biotrophic	Severity decreased at high CO ₂ with/without high O ₃	Decrease	Eastburn <i>et al.</i> (2010)
Soybean	Six necrotrophic pathogens	Necrotrophic	Decrease	Decrease	Boland <i>et al.</i> (2004)
Soybean	Stem canker – <i>Phytophthora sojae</i>	Necrotrophic	Increased phytoalexin glyceollin production at high CO ₂ in resistant but not in susceptible cultivar	Increase or decrease	Braga <i>et al.</i> (2006)
Soybean	2 necrotrophic pathogens	Necrotrophic	Increase	Increase	Boland <i>et al.</i> (2004)
Soybean	Brown spot – <i>Septoria glycines</i>	Necrotrophic	Increased at high CO ₂ with/without high O ₃ in 1 of 3 years	Increase	Eastburn <i>et al.</i> (2010)
Soybean	Sudden death syndrome – <i>Fusarium virguliforme</i>	Necrotrophic	No effect of elevated CO ₂ or O ₃	No change	Eastburn <i>et al.</i> (2010)
Wheat	Barley yellow dwarf virus	Biotrophic	Increase with climate change	Increase	Chakraborty <i>et al.</i> (1998)
Wheat	Two biotrophic pathogens	Biotrophic	Decrease under climate change	Decrease	Boland <i>et al.</i> (2004)
Wheat	Stripe rust – <i>Puccinia striiformis</i>	Biotrophic	Stimulation with small rise but inhibition at higher CO ₂ concentration	Increase or decrease	Manning & Tiedemann (1995)
Wheat	Stripe rust – <i>Puccinia striiformis</i>	Biotrophic	Increase with milder winter temperature	Increase	Coakley 1979
Wheat	Stripe rust – <i>Puccinia striiformis</i>	Biotrophic	Increase with pathogen adapting to higher temperature	Increase	Milus <i>et al.</i> (2006)
Wheat	Stripe rust – <i>Puccinia striiformis</i>	Biotrophic	Stripe rust severity increase with rising temperature	Increase	Chakraborty <i>et al.</i> (1998)
Wheat	Stripe rust – <i>Puccinia striiformis</i>	Biotrophic	Decrease with increased average annual temperature	Decrease	Yang <i>et al.</i> (1998)
Wheat	Stripe rust – <i>Puccinia graminis</i>	Biotrophic	Stimulation with small rise but inhibition at higher CO ₂ concentration	Increase or decrease	Manning & Tiedemann (1995)
Wheat	Leaf rust – <i>Puccinia triticina</i>	Biotrophic	Stimulation with small rise but inhibition at higher CO ₂ concentration	Increase or decrease	Manning & Tiedemann (1995)
Wheat	Powdery mildew – <i>Blumeria graminis</i>	Biotrophic	Increase or decrease based on plant nitrogen content	Increase or decrease	Thompson <i>et al.</i> (1993)
Wheat	Dwarf bunt – <i>Tilletia controversa</i>	Biotrophic	Increase under climate change	Increase	Boland <i>et al.</i> (2004)
Wheat	Four necrotrophic pathogens	Necrotrophic	Decrease under climate change	Decrease	Boland <i>et al.</i> (2004)
Wheat	Snow mould – <i>Fusarium</i> sp.	Necrotrophic	Enhanced snow mould high CO ₂	Increase	Manning & Tiedemann (1995)
Wheat	Spot blotch – <i>Cochliobolus sativus</i>	Necrotrophic	Increase under climate change	Increase	Sharma <i>et al.</i> (2007)
Wheat	Crown rot – <i>Fusarium pseudograminearum</i>	Necrotrophic	Increased inoculum at high CO ₂ , increased severity depending on cultivar and soil water	Increase	Chakraborty <i>et al.</i> (1998); Melloy <i>et al.</i> (2010)
Wheat	Head blight – <i>Fusarium</i> sp.	Necrotrophic	No effect of climate change	No change	Boland <i>et al.</i> (2004)

knowledge of the host, pathogen and disease epidemiology in a farming-systems context is essential for this, and since both parasitic and saprotrophic fitness will have to be considered for necrotrophs the list is considerably longer than for biotrophs. The retrospective analysis of long term-datasets such as herbarium specimens reviewed by Jeger & Pautasso (2008) and Fitt *et al.* (2011) will add to knowledge on the biology, distribution and adaptive responses of plant pathogens (and their vectors) to climate change. Quantitative data will increase the ability to counteract new risks posed by climate change for endemic pathogens and assist in the ability to circumvent any new introduction, potentially driven by climate-change events.

The ability to generate genetic variations as a response to changes in the environment enables plant pathogens to become highly adaptive organisms (Pangga *et al.*, 2011). New phenotypic strains or pathogenic races often arise in response to changing climate (Milus *et al.*, 2009) or the deployment of resistant host cultivars, such as with wheat rusts (Chakraborty *et al.*, 2010). Therefore, it is important to consider pathogen evolution and the effectiveness of resistant plant varieties for an accurate assessment of diseases on food crops in the future. It is also important to bear in mind that climate is only one driver of change when examining the future effects of plant disease on food crops. Other factors to consider include disease management practices such as the use of resistant cultivars and crop management practices such as time of planting, irrigation practices, stubble retention and crop rotations.

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